

Predictive e-diagnostics for multi-state movement
models: A theoretical framework for hidden
Markov models

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Abstract

Hidden Markov models are widely used to describe animal movement as a sequence of unobserved behavioural states generating observed step lengths, turning angles and other movement variables. Standard assessment workflows combine likelihood-based criteria, decoded state paths, simulation checks and graphical diagnostics. These tools are useful, but they do not by themselves give a sequentially valid account of adaptive model checking, nor do they distinguish clearly between observable predictive distributions and model-dependent reconstructions of latent states. This paper develops a theoretical framework for *predictive e-diagnostics* in multi-state movement models. A fitted HMM is treated as a sequential generator of held-out movement data. Diagnostic alternatives are encoded as predictable one-step or blockwise predictive densities, and the corresponding density ratios define e-value increments. The latent state process is integrated out by filtering, so the denominator is the observable predictive density, not a density conditional on a decoded path. Under a fixed train/validation protocol, the cumulative product of predictive density ratios is a nonnegative supermartingale under the fitted null generator and hence an e-process. The paper extends this basic guarantee to optional stopping, predictable switching among diagnostic alternatives, weighted and state-localized e-processes, feature-level diagnostics based on residual or circular-linear transformations, blockwise long-horizon diagnostics and conservative composite-null envelopes. These results provide a principled basis for movement-specific diagnostics targeting the number of states, angular misspecification, residual step-angle dependence, duration structure, temporal nonstationarity and long-term generative behaviour.

Keywords: Animal movement, hidden Markov model, e-value, e-process, sequential prediction, model diagnostics, filtering, circular data, anytime-valid inference

1 Introduction

Hidden Markov models (HMMs) are a central tool for analysing animal movement data when movement is represented as the observable output of an unobserved behavioural state process. In many applications, observations are constructed from locations as step lengths and turning angles, and latent states are interpreted as behavioural modes such as resting, encamped movement, foraging, exploratory movement or directed travel. Early movement models based on mixtures of random walks formalized this idea using step lengths and turning angles (Morales et al., 2004). HMM-type extensions are now standard for animal telemetry and movement ecology (Langrock et al., 2012; Zucchini et al., 2016), and software such as `moveHMM` (Michelot et al., 2016) and `momentuHMM` (McClintock and Michelot, 2018) has made these models accessible in applied ecological work.

Model assessment for movement HMMs often relies on AIC or BIC, likelihood comparisons, inspection of decoded state paths, residual checks, simulation-based checks and ecological interpretation of inferred behaviours. These practices are useful, but they raise at least three statistical issues. First, diagnostics are often chosen after seeing the data, which makes the inferential status of repeated exploratory checking unclear. Second, the latent states are sometimes treated as if they were observed after decoding, even though decoded states remain model-dependent reconstructions. Third, a high fitted likelihood or a plausible state interpretation does not ensure that the model is a good sequential generator of future movement patterns, especially for long trajectories, held-out individuals, directional persistence, state dwell times or spatially aggregated summaries.

This paper proposes a predictive alternative. The fitted HMM is evaluated as a sequential generator of a validation trajectory. At each time t , the null model provides an observable predictive density

$$p_0(y_t \mid \mathcal{F}_{t-1}).$$

A diagnostic alternative provides another predictive density

$$q_t(y_t \mid \mathcal{F}_{t-1}),$$

chosen before Y_t is observed. The ratio

$$E_t = \frac{q_t(Y_t \mid \mathcal{F}_{t-1})}{p_0(Y_t \mid \mathcal{F}_{t-1})}$$

is a predictive e-value increment. Its cumulative product is an e-process under the null generator. The resulting evidence process can be monitored over time and compared with thresholds of the form $1/\alpha$, with the usual anytime-valid interpretation supplied by Ville's inequality (Ville, 1939; Howard et al., 2020). The construction connects movement-model diagnostics to the prequential view of statistical evaluation (Dawid, 1984) and to the modern literature on e-values, safe testing and anytime-valid inference (Vovk and Wang, 2021; Grünwald et al., 2024).

The main point is not simply to replace likelihood ratios by a new terminology. The contribution is to treat movement-model validation as a predictable sequence of diagnostic bets against a fitted generator. This leads to several methodological consequences. The denominator must be the observable predictive density, obtained by filtering over latent states. Diagnostic alternatives may be switched or mixed predictably. Local evidence can be accumulated only over selected periods, habitats or

139 soft state weights while preserving e-process validity. Movement-specific summaries,
140 including circular residuals, step-angle dependence measures and blockwise spatial
141 summaries, can be converted into feature-level e-values whenever their null predictive
142 law is available or can be estimated on independent data.
143

144
145 The paper is written as a theoretical setup for a first methodological article. The
146 core guarantees are exact under a fixed train/validation protocol. This means that the
147 null model and diagnostic alternatives are fitted or specified before validation, and the
148 validation sequence is then used only to compute predictive ratios. The same-sample
149 workflow in which the analyst fits the model, chooses diagnostics and evaluates them
150 all on the same observations is not covered by the exact guarantees below unless
151 additional sample splitting, cross-fitting or pre-specified conditioning arguments are
152 used.
153

154 The numerical section is deliberately focused. The main article reports the sce-
155 narios that support the core methodological message: calibration under a fixed null
156 generator, targeted movement diagnostics, predictable localization and predictable
157 diagnostic combination. Additional sensitivity checks and extensions are reported in
158 the supplementary material.
159

160 ***Main contributions.***

161 The paper develops the following theoretical components.

- 162 (C1) A predictive e-process theorem for fixed movement HMMs evaluated on held-out
163 validation sequences.
- 164 (C2) A filtering result showing that validity is based on marginal observable prediction,
165 not on densities conditional on decoded state paths.
- 166 (C3) Optional-stopping, predictable-switching and mixture results for diagnostic
167 alternatives.
- 168 (C4) Weighted and state-localized e-processes that permit valid localization of model
169 failure by time, habitat or predicted behavioural state.

(C5) Feature-level and blockwise e-diagnostics for circular-linear movement summaries	185
and long-horizon generative checks.	186
	187
(C6) A clear separation between exact split-sample validity, exploratory same-sample	188
diagnostics and conservative composite-null extensions.	189
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2 Validation protocol and movement HMM

2.1 Training information and validation filtration

Let $(\Omega, \mathcal{A}, \mathbb{P}_0)$ be the underlying probability space under the fitted null generator.

Let $\mathcal{H} \subseteq \mathcal{A}$ be the pre-validation sigma-field generated by the training data, fitted parameters, preprocessing choices, random seeds, and diagnostic-menu choices. All modelling choices in $\mathcal{H}$ must be fixed before validation. The validation trajectory is indexed by $t = 1, \dots, T$. The observed movement variable is a measurable mapping $Y_t : \Omega \rightarrow \mathcal{Y}_t$ given by

$$Y_t = (L_t, \Theta_t), \quad (1)$$

where L_t is a step length and Θ_t is a turning angle. More generally, Y_t may include several movement streams or covariates observed at the same temporal resolution. The latent behavioural state is

$$S_t \in \{1, \dots, K\}. \quad (2)$$

The observable validation filtration is the completion of

$$\mathcal{F}_t = \sigma(\mathcal{H}, Y_1, \dots, Y_t), \quad t \geq 0, \quad (3)$$

Notation	Meaning
Y_t	Observed movement variable at time t , often (L_t, Θ_t)
L_t	Step length at time t
Θ_t	Turning angle at time t
S_t	Latent behavioural state at time t
K	Number of states in the null HMM
$a_{1 0}(s)$	Initial predictive probability for state s
$\mathcal{H}$	Training information fixed before validation
$\mathcal{F}_t$	Observable validation filtration, $\sigma(\mathcal{H}, Y_1, \dots, Y_t)$
M_0	Null HMM or fitted null generator
$p_0(y_t \mathcal{F}_{t-1})$	Observable predictive density under M_0
$q_t(y_t \mathcal{F}_{t-1})$	Diagnostic predictive density or subdensity at time t
E_t	Predictive e-value increment at time t
$E_{1:t}$	Cumulative product of increments up to time t
α	Anytime-valid monitoring level

Table 1 Notation used throughout the theoretical setup.

where $\mathcal{F}_0 = \mathcal{H}$. A diagnostic rule is called *predictable* if the object used at time t is $\mathcal{F}_{t-1}$ -measurable. Informally, predictable means that the diagnostic bet for Y_t is chosen before Y_t is revealed.

2.2 Null HMM

Conditional on $\mathcal{H}$, the null model M_0 is a fixed K -state HMM with initial distribution π_0 , transition matrix $\mathbf{\Gamma}_0$ and state-dependent observation densities. Write

$$f_{0,s}(y_t) = p_0(y_t | S_t = s, \mathcal{F}_{t-1}^{\text{cov}}) \quad (4)$$

for the state-dependent density in state s , where $\mathcal{F}_{t-1}^{\text{cov}}$ denotes any covariate information allowed in the fitted model and known before Y_t . To simplify notation, covariates are suppressed below. In a common movement specification,

$$f_{0,s}(l_t, \theta_t) = f_{0,s}^L(l_t) f_{0,s}^\Theta(\theta_t), \quad (5)$$

for example with Gamma step lengths and von Mises turning angles. The factorization in (5) is a modelling assumption, not a requirement of the e-process construction.

Target	Validity claim	Practical interpretation
Fixed fitted generator	Exact conditional e-process	“Does this fitted HMM generate the validation trajectory?”
True parametric HMM with estimated parameters	Not exact without adjustment	Requires parameter uncertainty treatment
Composite family	Uniform e-process via envelope or other safe construction	Conservative but genuine family-wise null

Table 2 Distinct statistical targets for movement model validation. This paper focuses primarily on the fixed fitted generator, treating the diagnostic as a check of the specific predictive model resulting from training.

The HMM formulation also allows covariate-dependent transition probabilities, time-varying state-dependent parameters and multivariate observations, provided the one-step predictive density is well-defined and the covariates used for prediction are fixed before observing Y_t .

3 Observable predictive density by filtering

The central object is the one-step-ahead observable predictive density. Since S_t is latent, this density must marginalize over the predictive distribution of the state:

$$p_0(y_t | \mathcal{F}_{t-1}) = \sum_{s=1}^K f_{0,s}(y_t) \mathbb{P}_0(S_t = s | \mathcal{F}_{t-1}). \quad (6)$$

This is the density of the next observed movement variable under the fitted HMM, conditional only on the observed past and the training information.

Let

$$a_{t|t-1}(s) = \mathbb{P}_0(S_t = s | \mathcal{F}_{t-1}) \quad (7)$$

denote the predictive state probabilities before observing Y_t . This recursion must be explicitly initialized using the stationary or fitted initial distribution, $a_{1|0}(s) = \pi_0(s)$. In applications with multiple validation individuals (such as cross-validation folds), the filter must be restarted for each held-out individual i using $a_{i,1|0}(s) = \pi_{0,i}(s)$.

323 Then

$$324 \quad p_0(y_t \mid \mathcal{F}_{t-1}) = \sum_{s=1}^K a_{t|t-1}(s) f_{0,s}(y_t). \quad (8)$$

327 After observing $Y_t = y_t$, the filtered probabilities are

$$328 \quad a_{t|t}(s) = \frac{a_{t|t-1}(s) f_{0,s}(y_t)}{\sum_{r=1}^K a_{t|t-1}(r) f_{0,r}(y_t)}. \quad (9)$$

333 The next predictive state probabilities are

$$334 \quad a_{t+1|t}(s) = \sum_{r=1}^K a_{t|t}(r) \Gamma_{0,t}(r, s), \quad (10)$$

340 where $\Gamma_{0,t}$ denotes the transition probabilities. If transition probabilities depend on
 341 covariates, $\Gamma_{0,t}$ must be $\mathcal{F}_t$ -measurable (i.e., any covariates used for predicting S_{t+1}
 342 must be known at time t). Crucially, spatial covariates computed from the animal's
 343 arrival location (such as habitat at the end of a step or distance to water after moving)
 344 are only predictable if they can be determined exactly from information available
 345 before observing Y_t , or if the diagnostic explicitly targets the joint distribution of the
 346 movement and the spatial context. The time-homogeneous HMM is the special case
 347 where $\Gamma_{0,t} \equiv \Gamma_0$. Equations (6) to (10) provide the computational basis of the null
 348 predictive density. They are also the reason the method differs from diagnostics based
 349 on Viterbi or other decoded state paths.

359 4 Predictive e-values

362 4.1 Diagnostic alternatives as predictable densities

364 A diagnostic alternative is represented by a predictable Markov kernel or sub-Markov
 365 kernel Q_t . When represented by a density or subdensity $q_t(y \mid \mathcal{F}_{t-1})$ with respect
 366 to a common dominating measure, the joint map $(\omega, y) \mapsto q_t(y \mid \mathcal{F}_{t-1})(\omega)$ must be

$\mathcal{F}_{t-1} \otimes \mathcal{Y}$ -measurable. The subscript t allows the diagnostic to depend on the observed past. The key requirement is predictability: the rule defining q_t must be fixed before observing Y_t .

Examples include:

- a $K + 1$ state HMM used to diagnose whether the null HMM has too few states;
- a more flexible angular distribution used to diagnose misspecified turning angles;
- a joint step-angle model used to diagnose residual dependence between L_t and Θ_t within behavioural states;
- a duration-aware alternative, such as a hidden semi-Markov model (HSMM), used to diagnose non-geometric state dwell times;
- a flexible predictive model trained on independent data and used as a diagnostic competitor to the fitted HMM.

These alternatives are diagnostic bets. They need not be interpreted as true data-generating mechanisms.

Throughout, all predictive distributions and diagnostic bets are represented by densities with respect to a common dominating measure μ_t on the observation space at time t . This can be Lebesgue measure for continuous step-angle observations, counting measure for discrete observations or a product measure for mixed data. The convention is that ratios are evaluated only on the support of the null predictive density; outside that support the null event has probability zero, and we adopt the support convention that $q_t(y \mid \mathcal{F}_{t-1}) = 0$ whenever $p_0(y \mid \mathcal{F}_{t-1}) = 0$. Diagnostic bets q_t may be fixed before validation, or updated sequentially online using past validation observations, provided the update rule is predictable.

Definition 1 (Predictive e-value increment) For $t = 1, \dots, T$, define

$$E_t = \frac{q_t(Y_t \mid \mathcal{F}_{t-1})}{p_0(Y_t \mid \mathcal{F}_{t-1})}, \quad (11)$$

whenever the ratio is well-defined under M_0 .

415 **Definition 2** (Predictive e-process) The cumulative process is

$$416 \quad E_{1:t} = \prod_{u=1}^t E_u, \quad E_{1:0} = 1. \quad (12)$$

419 Equivalently,

$$421 \quad \log E_{1:t} = \sum_{u=1}^t \{\log q_u(Y_u \mid \mathcal{F}_{u-1}) - \log p_0(Y_u \mid \mathcal{F}_{u-1})\}. \quad (13)$$

425 4.2 Main guarantee

427 **Theorem 1** (Predictive e-process for a fixed HMM) *Assume that, conditional on $\mathcal{H}$, the*
 428 *validation sequence is generated under the null HMM M_0 , and that $p_0(y \mid \mathcal{F}_{t-1})$ is the*
 429 *corresponding observable predictive density with respect to μ_t . Let $q_t(y \mid \mathcal{F}_{t-1})$ be an $\mathcal{F}_{t-1}$ -*
 430 *measurable diagnostic subdensity satisfying*

$$433 \quad \int q_t(y \mid \mathcal{F}_{t-1}) d\mu_t(y) \leq 1 \quad (14)$$

436 *almost surely. Then*

$$437 \quad \mathbb{E}_0(E_t \mid \mathcal{F}_{t-1}) \leq 1. \quad (15)$$

439 *Consequently, $(E_{1:t})_{0 \leq t \leq T}$ is a nonnegative supermartingale under M_0 . If $q_t(\cdot \mid \mathcal{F}_{t-1})$ is*
 440 *a density assigning all its mass to the support of $p_0(\cdot \mid \mathcal{F}_{t-1})$, then equality holds and*
 441 *$(E_{1:t})_{0 \leq t \leq T}$ is a martingale.*

446 *Proof* Conditional on $\mathcal{F}_{t-1}$, Y_t has density $p_0(y \mid \mathcal{F}_{t-1})$ under M_0 . Therefore,

$$448 \quad \mathbb{E}_0(E_t \mid \mathcal{F}_{t-1}) = \int_{\{p_0 > 0\}} \frac{q_t(y \mid \mathcal{F}_{t-1})}{p_0(y \mid \mathcal{F}_{t-1})} p_0(y \mid \mathcal{F}_{t-1}) d\mu_t(y) \quad (16)$$

$$451 \quad = \int_{\{p_0 > 0\}} q_t(y \mid \mathcal{F}_{t-1}) d\mu_t(y) \quad (17)$$

$$453 \quad \leq 1. \quad (18)$$

456 For the cumulative process,

$$458 \quad \mathbb{E}_0(E_{1:t} \mid \mathcal{F}_{t-1}) = E_{1:t-1} \mathbb{E}_0(E_t \mid \mathcal{F}_{t-1}) \quad (19)$$

$$459 \quad \leq E_{1:t-1}. \quad (20)$$

Thus $(E_{1:t})_{0 \leq t \leq T}$ is a nonnegative supermartingale. If q_t is a density supported by the null support, the conditional expectation is one at every time, giving the martingale case. $\square$

Corollary 2 (Optional stopping and anytime-valid monitoring) *Under the conditions of Theorem 1, for any $\alpha \in (0, 1)$,*

$$\mathbb{P}_0 \left(\sup_{0 \leq t \leq T} E_{1:t} \geq \frac{1}{\alpha} \right) \leq \alpha. \quad (21)$$

Equivalently,

$$\mathbb{P}_0 \left(\sup_{0 \leq t \leq T} \log E_{1:t} \geq \log(1/\alpha) \right) \leq \alpha. \quad (22)$$

Moreover, for any stopping time τ with respect to $(\mathcal{F}_t)$, the stopped process $E_{1:\tau \wedge T}$ is an e-value under M_0 , meaning $\mathbb{E}_0(E_{1:\tau \wedge T}) \leq 1$.

Proof Ville's maximal inequality applied to the nonnegative supermartingale $(E_{1:t})_{0 \leq t \leq T}$ gives the first display. The logarithmic version follows from monotonicity of the logarithm. For a bounded stopping time τ , optional stopping for nonnegative supermartingales gives $\mathbb{E}_0(E_{1:\tau}) \leq E_{1:0} = 1$. The general case follows by applying the bounded result to $\tau \wedge m$ and using Fatou's lemma as $m \rightarrow \infty$. $\square$

Corollary 3 (Anytime p-value induced by an e-process) *Define*

$$p_t^{\text{any}} = \min \left\{ 1, \frac{1}{\sup_{0 \leq u \leq t} E_{1:u}} \right\}. \quad (23)$$

Then p_t^{any} is an anytime-valid p-value process in the sense that, under M_0 ,

$$\mathbb{P}_0 \left(\exists 0 \leq t \leq T : p_t^{\text{any}} \leq \alpha \right) \leq \alpha. \quad (24)$$

Proof The event $\{\exists 0 \leq t \leq T : p_t^{\text{any}} \leq \alpha\}$ is the same as the event $\{\sup_{0 \leq t \leq T} E_{1:t} \geq 1/\alpha\}$, which is controlled by Corollary 2. $\square$

507 5 Why filtering is essential

508

509 The proof of Theorem 1 uses only one fact about the null model: the denominator must
 510 be the true conditional density of the next observable variable given the observable
 511 past. In an HMM, that density is not obtained by conditioning on a decoded state. It
 512 is obtained by marginalizing over the predictive state distribution.
 513
 514
 515

516

517 **Proposition 4** (Validity is based on marginal observable prediction) *For an HMM, the*
 518 *denominator in (11) is the marginal observable predictive density*

519

$$520 \quad p_0(y_t \mid \mathcal{F}_{t-1}) = \sum_{s=1}^K f_{0,s}(y_t) \mathbb{P}_0(S_t = s \mid \mathcal{F}_{t-1}). \quad (25)$$

521

522 *The e-process validity does not require observing S_t and does not generally follow from*
 523 *replacing S_t by a decoded path $\hat{S}_t$.*

524

525

526 *Proof* The proof of Theorem 1 requires the denominator to be the conditional density of Y_t
 527 given $\mathcal{F}_{t-1}$. In an HMM this conditional law is obtained by summing over the unobserved
 528 state S_t using the predictive state probabilities. A decoded path $\hat{S}_t$ is a statistic produced
 529 by a fitted model and observed data. It does not appear in the conditional law of Y_t given
 530 $\mathcal{F}_{t-1}$ unless the decoded state is known to equal the true latent state, which is not the usual
 531 HMM setting. □

532

533

534 *Example 1* (A state-conditioned denominator can violate e-validity) Consider a two-state
 535 model at a single validation time with predictive state probabilities 1/2 and 1/2. Let

536

$$537 \quad f_1(y) = \phi(y; -m, 1), \quad f_2(y) = \phi(y; m, 1),$$

538

539 where $\phi(\cdot; \mu, 1)$ is the $N(\mu, 1)$ density and $m > 0$. The observable null density is

540

$$541 \quad p_0(y) = \frac{1}{2} f_1(y) + \frac{1}{2} f_2(y).$$

542

543 Suppose one incorrectly uses $f_1(y)$ as the denominator because a decoded or assigned state
 544 is set to state 1, and suppose the diagnostic numerator is $q(y) = f_2(y)$. The expected ratio

545

under the true observable null is

$$\mathbb{E}_0 \left\{ \frac{f_2(Y)}{f_1(Y)} \right\} = \int \frac{f_2(y)}{f_1(y)} \left\{ \frac{1}{2} f_1(y) + \frac{1}{2} f_2(y) \right\} dy \quad (26)$$

$$= \frac{1}{2} + \frac{1}{2} \int \frac{f_2(y)^2}{f_1(y)} dy \quad (27)$$

$$= \frac{1}{2} + \frac{1}{2} \exp(4m^2), \quad (28)$$

which is larger than 1. Thus the ratio is not an e-value under the observable null. The problem is not the use of likelihood ratios; the problem is the use of the wrong denominator.

In movement terms, replacing the filtering mixture by a Viterbi state-specific turning-angle density can turn a descriptive behavioural state label into an invalid predictive denominator.

Remark 1 (Decoded states are useful after, not before, validation) A Viterbi path, local decoding or posterior state classification may be useful for interpretation after a diagnostic signal has been found. For example, one may report whether large local log-increments occur during times with high smoothed probability of a particular behaviour. Such summaries are descriptive unless the weights used for localization are predictable. Section 7 gives a valid predictable alternative based on filtered state probabilities.

6 Predictable diagnostic selection, mixtures and parallel diagnostics

In practice, the analyst may not want to commit to a single diagnostic alternative. One may want to compare a $K + 1$ state HMM, an angular alternative, a copula alternative and a duration alternative. E-values are convenient here, but care is needed. Predictable switching and mixtures are valid. Products of parallel diagnostics computed on the same validation observations are not automatically valid.

599 **Proposition 5** (Predictable mixtures of diagnostic alternatives) *Let $q_{j,t}(y \mid \mathcal{F}_{t-1})$, $j =$
600 $1, \dots, J$, be predictable diagnostic subdensities satisfying*

$$602 \int q_{j,t}(y \mid \mathcal{F}_{t-1}) \mu_t(dy) \leq 1.$$

603
604 Let $w_{j,t} \geq 0$ be predictable weights with $\sum_{j=1}^J w_{j,t} \leq 1$. Define

$$605 q_t^{\text{mix}}(y \mid \mathcal{F}_{t-1}) = \sum_{j=1}^J w_{j,t} q_{j,t}(y \mid \mathcal{F}_{t-1}) \quad (29)$$

606
607 and

$$608 E_t^{\text{mix}} = \frac{q_t^{\text{mix}}(Y_t \mid \mathcal{F}_{t-1})}{p_0(Y_t \mid \mathcal{F}_{t-1})}. \quad (30)$$

609
610 Then E_t^{mix} is a conditional e-value increment, and $\prod_{u \leq t} E_u^{\text{mix}}$ is an e-process under M_0 .

611
612
613 *Proof* The mixture q_t^{mix} is predictable and integrates to at most $\sum_j w_{j,t} \leq 1$. The result
614 follows directly from Theorem 1. $\square$

615
616
617 **Corollary 6** (Predictable switching) *Let $J_t \in \{1, \dots, J\}$ be $\mathcal{F}_{t-1}$ -measurable. Then*

$$618 E_t^{\text{switch}} = \frac{q_{J_t,t}(Y_t \mid \mathcal{F}_{t-1})}{p_0(Y_t \mid \mathcal{F}_{t-1})} \quad (31)$$

619
620 is a conditional e-value increment. Thus the analyst may switch among a pre-specified menu of
621 diagnostic alternatives using only information available before the next validation observation.

622
623 *Proof* This is Proposition 5 with $w_{j,t} = \mathbf{1}\{J_t = j\}$. $\square$

624
625
626 **Proposition 7** (Parallel diagnostics on the same validation data) *Suppose $E^{(1)}, \dots, E^{(J)}$
627 are e-values under M_0 computed from the same validation experiment, so that $\mathbb{E}_0 E^{(j)} \leq 1$
628 for each j . If $w_j \geq 0$ and $\sum_j w_j \leq 1$, then*

$$629 E^{\text{avg}} = \sum_{j=1}^J w_j E^{(j)} \quad (32)$$

is also an e-value under M_0 . Consequently,

$$\mathbb{P}_0 \left(\max_{1 \leq j \leq J} w_j E^{(j)} \geq \frac{1}{\alpha} \right) \leq \alpha. \quad (33)$$

In contrast, the product $\prod_{j=1}^J E^{(j)}$ is not generally an e-value when the diagnostics are computed in parallel on the same observations.

Proof Linearity of expectation gives

$$\mathbb{E}_0 E^{\text{avg}} = \sum_j w_j \mathbb{E}_0 E^{(j)} \leq \sum_j w_j \leq 1.$$

If $\max_j w_j E^{(j)} \geq 1/\alpha$, then $E^{\text{avg}} \geq 1/\alpha$, so Markov's inequality applied to the e-value E^{avg} gives the displayed bound. The final statement is a warning: without additional conditional independence or a sequential construction using fresh information, products of e-values need not have expectation at most one. $\square$

Remark 2 (Recommended reporting for several diagnostics) For a diagnostic menu evaluated on the same validation sequence, report the individual e-processes and either use a weighted average e-value or pre-assigned weighted thresholds. A product of diagnostic e-processes is valid only when the factors are accumulated as a sequential process with each new factor conditionally valid given all previous diagnostic outcomes and using the appropriate next unit of information.

7 Localization, state weighting and tempering

A global e-process answers whether a diagnostic alternative accumulates evidence against the null generator. Movement ecologists usually also want to know where the failure occurs: during migration, in a given habitat, for one individual, or during time periods in which the fitted model predicts a particular behavioural state. The following result gives a valid way to localize evidence when the localization weights are predictable.

691 **Proposition 8** (Predictable weighted localization) *Let E_t be any conditional e-value incre-*
692 *ment satisfying $\mathbb{E}_0(E_t \mid \mathcal{F}_{t-1}) \leq 1$. Let W_t be any $\mathcal{F}_{t-1}$ -measurable weight with $0 \leq W_t \leq 1$.*
693

694 *Define*

$$695 \quad E_t^{(W)} = 1 + W_t(E_t - 1). \quad (34)$$

697 *Then $E_t^{(W)} \geq 0$ and*

$$698 \quad \mathbb{E}_0(E_t^{(W)} \mid \mathcal{F}_{t-1}) \leq 1. \quad (35)$$

699
700 *Consequently, $\prod_{u \leq t} E_u^{(W)}$ is an e-process under M_0 .*
701
702
703
704

705 *Proof* Since $E_t \geq 0$ and $0 \leq W_t \leq 1$, $E_t^{(W)} = (1 - W_t) + W_t E_t \geq 0$. Also,

$$708 \quad \mathbb{E}_0(E_t^{(W)} \mid \mathcal{F}_{t-1}) = 1 + W_t\{\mathbb{E}_0(E_t \mid \mathcal{F}_{t-1}) - 1\} \quad (36)$$

$$709 \quad \leq 1. \quad (37)$$

711
712 The product result follows from the same supermartingale argument as in Theorem 1. $\square$
713
714

715 **Corollary 9** (Segment-specific e-processes) *If B_t is a predictable indicator of a biologically*
716 *meaningful segment, such as a known season, a habitat class known before observing the*
717 *movement response, or a held-out individual, then*
718

$$719 \quad E_t^{(B)} = \begin{cases} E_t, & B_t = 1, \\ 1, & B_t = 0, \end{cases} \quad (38)$$

724 *yields a valid segment-specific e-process. Thus evidence can be accumulated only over a pre-*
725 *specified or predictably selected subset of times.*
726
727
728
729

730 *Proof* This is Proposition 8 with $W_t = B_t \in \{0, 1\}$. $\square$
731
732

733 **Corollary 10** (Soft state-localized e-processes) *For a fitted HMM, set*

$$735 \quad W_t^{(s)} = a_{t|t-1}(s) = \mathbb{P}_0(S_t = s \mid \mathcal{F}_{t-1}). \quad (39)$$

736

Then

$$E_t^{(s)} = 1 + a_{t|t-1}(s)(E_t - 1) \quad (40)$$

defines a valid e-process localized to times at which state s is predicted to be likely under the null model.

Proof The filtered predictive probability $a_{t|t-1}(s)$ is $\mathcal{F}_{t-1}$ -measurable and lies in $[0, 1]$, so Proposition 8 applies. $\square$

Proposition 11 (Power-tempered and linearly tempered e-increments) *Let E_t be a conditional e-value increment. If $\lambda_t \in [0, 1]$ is predictable, then both*

$$\tilde{E}_t^{\text{pow}} = E_t^{\lambda_t} \quad (41)$$

(with the convention $0^0 = 1$) and

$$\tilde{E}_t^{\text{lin}} = 1 + \lambda_t(E_t - 1) \quad (42)$$

are conditional e-value increments. Their products are e-processes.

Proof The linear case follows from Proposition 8 with $W_t = \lambda_t$. For the power case, Jensen's inequality applied to the concave function $x \mapsto x^{\lambda_t}$ gives

$$\mathbb{E}_0(E_t^{\lambda_t} \mid \mathcal{F}_{t-1}) \leq \{\mathbb{E}_0(E_t \mid \mathcal{F}_{t-1})\}^{\lambda_t} \leq 1.$$

$\square$

Remark 3 (Interpretation of localized evidence) The process in Corollary 10 is not the same as conditioning on the event that the true latent state equals s . It is a valid soft localization based on the null model's predictive probability of state s . Smoothed state probabilities using future observations can be useful for visualization, but they are not predictable and therefore do not by themselves preserve the e-process guarantee.

8 Feature-level and blockwise e-diagnostics

The previous sections use full predictive densities for Y_t . In movement applications, the diagnostically relevant object may be a lower-dimensional feature, such as a circular residual, a transformed step length, a copula residual, a barrier-crossing indicator or a blockwise spatial statistic. E-values can be constructed directly from such features if their null conditional law is known.

Theorem 12 (Feature-level predictive e-values) *Let $Z_t = \psi_t(Y_t, \mathcal{F}_{t-1})$ be a diagnostic feature whose transformation rule ψ_t is predictable. Let $g_{0,t}(z | \mathcal{F}_{t-1})$ be the conditional density of Z_t under M_0 , and let $g_{1,t}(z | \mathcal{F}_{t-1})$ be a predictable diagnostic subdensity satisfying $\int g_{1,t}(z | \mathcal{F}_{t-1}) \nu_t(dz) \leq 1$. Define*

$$R_t = \frac{g_{1,t}(Z_t | \mathcal{F}_{t-1})}{g_{0,t}(Z_t | \mathcal{F}_{t-1})}. \quad (43)$$

Then R_t is a conditional e-value increment under M_0 , and $\prod_{u \leq t} R_u$ is an e-process.

Proof Conditional on $\mathcal{F}_{t-1}$, Z_t has density $g_{0,t}(z | \mathcal{F}_{t-1})$ under M_0 . Therefore,

$$\mathbb{E}_0(R_t | \mathcal{F}_{t-1}) = \int_{\{g_0 > 0\}} g_{1,t}(z | \mathcal{F}_{t-1}) \nu_t(dz) \leq 1.$$

The product result follows as before. $\square$

8.1 Rosenblatt residuals and step-angle dependence

Feature-level e-values are especially useful for circular-linear diagnostics. Suppose the null model gives a joint predictive distribution for $Y_t = (L_t, \Theta_t)$. A sequential Rosenblatt transform can be defined by

$$U_t^L = F_{0,t}^L(L_t | \mathcal{F}_{t-1}), \quad (44)$$

$$U_t^\Theta = F_{0,t}^{\Theta|L}(\Theta_t | L_t, \mathcal{F}_{t-1}), \quad (45)$$

where the second conditional distribution respects the circular support of Θ_t . Under a continuous correctly specified null predictive distribution, (U_t^L, U_t^Θ) is conditionally uniform on $[0, 1]^2$. A copula diagnostic can therefore use

$$R_t^{\text{cop}} = c_\rho(U_t^L, U_t^\Theta), \quad (46)$$

where c_ρ is a copula density on $[0, 1]^2$, with ρ fixed from training or chosen predictably. Theorem 12 gives conditional e-validity because c_ρ integrates to one under the uniform null. This construction targets residual step-angle dependence after the fitted HMM has already accounted for latent behavioural states.

Remark 4 (Circular data) Turning angles live on the circle, so residual diagnostics should respect periodicity. One may work with circular distribution functions relative to a chosen origin, with probability integral transforms for von Mises or mixture von Mises predictive distributions, or with features such as $\cos(\Theta_t - \mu_t)$ whose null law is derived from the circular predictive distribution. The e-value requirement is not normality of residuals; it is knowledge or independent estimation of the feature's null predictive law.

8.2 Blockwise diagnostics for long-horizon movement summaries

One-step predictive diagnostics are local. Movement-model failures can also appear only over longer horizons, for example through utilization distributions, mean squared displacement, straightness, return times, residence times or barrier crossing frequencies. These can be handled by blockwise e-values.

Theorem 13 (Blockwise predictive e-values) *Let $0 = \tau_0 < \tau_1 < \tau_2 < \dots$ be stopping times such that τ_j is finite and the next block rule is determined before the block begins. For block*

875 j , write

$$876 \quad Y_{B_j} = Y_{\tau_{j-1}+1:\tau_j}. \quad (47)$$

878 Let $p_{0,j}(y_{B_j} \mid \mathcal{F}_{\tau_{j-1}})$ be the conditional null predictive density of the stopped block on the
 879 variable-length product space $\cup_{m=1}^{\infty} \mathcal{Y}^m$, and let $q_j(y_{B_j} \mid \mathcal{F}_{\tau_{j-1}})$ be a predictable diagnostic
 880 subdensity on the same space. Define

$$881 \quad E_j^{\text{block}} = \frac{q_j(Y_{B_j} \mid \mathcal{F}_{\tau_{j-1}})}{p_{0,j}(Y_{B_j} \mid \mathcal{F}_{\tau_{j-1}})}. \quad (48)$$

883 Then $(\prod_{r \leq j} E_r^{\text{block}})_{j \geq 0}$ is an e-process under M_0 .

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 886 *Proof* Conditional on $\mathcal{F}_{\tau_{j-1}}$, the block Y_{B_j} has density $p_{0,j}$ under M_0 . The same density-ratio
 887 calculation as in Theorem 1 gives

$$888 \quad \mathbb{E}_0(E_j^{\text{block}} \mid \mathcal{F}_{\tau_{j-1}}) \leq 1.$$

889 The cumulative product over blocks is therefore a nonnegative supermartingale with respect
 890 to the block filtration $(\mathcal{F}_{\tau_j})_{0 \leq j \leq J}$. $\square$

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 893 **Corollary 14** (Feature-level block diagnostics) Let $Z_j = h_j(Y_{B_j}, \mathcal{F}_{\tau_{j-1}})$ be a predictable
 894 block feature such as a barrier-crossing indicator, a displacement statistic or a summary of
 895 space use over the block. If the conditional null density or mass function of Z_j is $g_{0,j}$ and a
 896 diagnostic alternative is $g_{1,j}$, then $g_{1,j}(Z_j)/g_{0,j}(Z_j)$ is a blockwise e-value increment.

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 899 *Proof* Apply Theorem 12 at the block level. $\square$

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 901 *Remark 5* (Simulation-based block laws) When the null law of a block feature is unavailable
 902 analytically, it may be approximated by simulation from the fitted HMM using only training
 903 information and the observed block covariates. Exact finite-sample e-validity then depends on
 904 how the density or mass estimate is constructed. A conservative recommendation for a first
 905 implementation is to fit any feature-level diagnostic law on independent null simulations or
 906 training data, and to reserve a separate validation trajectory for the final e-process calculation.

9 Estimated parameters, split validation and composite nulls

The exact theory above treats M_0 as a fixed generator at validation time. In applied movement modelling, M_0 is fitted. This section separates three interpretations: conditional validity for a fitted generator, cross-fitted e-values over independent validation units and conservative composite-null envelopes.

9.1 Conditional validity after training

Theorem 15 (Split-sample conditional validity) *Let $\mathcal{H}$ contain the training data and all fitted parameters of the null and diagnostic alternatives. Suppose that, conditional on $\mathcal{H}$, the validation trajectory is generated from the fitted null predictive law $p_0(\cdot \mid \mathcal{F}_{t-1})$, and suppose q_t is predictable with respect to $\mathcal{F}_{t-1} = \mathcal{H} \vee \sigma(Y_1, \dots, Y_{t-1})$. Then the process in Theorem 1 is an e-process conditional on $\mathcal{H}$. In particular,*

$$\mathbb{P}_0 \left(\sup_{0 \leq t \leq T} E_{1:t} \geq 1/\alpha \mid \mathcal{H} \right) \leq \alpha \quad (49)$$

almost surely, and the same bound holds unconditionally.

Proof Conditional on $\mathcal{H}$, the fitted densities are fixed and the validation filtration satisfies the assumptions of Theorem 1. The conditional bound follows from Corollary 2. Taking expectations over $\mathcal{H}$ gives the unconditional bound. $\square$

This theorem clarifies the interpretation of the first article’s target: the exact null is that the held-out validation sequence is generated by the fitted HMM. It is a diagnostic check of the fitted generator, not a claim that the parameter estimator has no uncertainty.

9.2 Independent validation units and cross-fitted averages

A natural movement design trains on some individuals and validates on held-out individuals. If each held-out individual yields an e-value, these e-values can be reported individually or combined by averaging.

Proposition 16 (Cross-fitted average over validation units) *Suppose $E^{(i)}$ is an e-value under M_0 for validation unit i , for $i = 1, \dots, n$, where each $E^{(i)}$ may be constructed using a model fitted on data not including the validation responses of unit i . If $w_i \geq 0$ and $\sum_i w_i \leq 1$, then*

$$E^{\text{cf}} = \sum_{i=1}^n w_i E^{(i)} \quad (50)$$

is an e-value under M_0 .

Proof The proof is the same as Proposition 7: $\mathbb{E}_0 E^{\text{cf}} = \sum_i w_i \mathbb{E}_0 E^{(i)} \leq 1$. No independence among the e-values is needed for averaging. Independence or a suitable sequential conditional construction would be needed for products. $\square$

Remark 6 (Leave-one-individual-out products) A leave-one-individual-out workflow may produce individually valid e-values, but the product across individuals is not automatically valid because the fitted models may use overlapping training sets and the e-values are computed in parallel. The weighted average in Proposition 16 is a safer default unless a sequential data-collection or sample-splitting design is used.

9.3 A conservative envelope for composite null families

The fitted-generator null is often the most transparent target for model criticism. If one instead wants a single process valid for every parameter value in a null family, a conservative envelope construction is possible.

Proposition 17 (Composite-null envelope) *Let $\{p_\theta(y \mid \mathcal{F}_{t-1}) : \theta \in \Theta_0\}$ be a family of null predictive densities. Assume that the pointwise supremum*

$$\bar{p}_0(y \mid \mathcal{F}_{t-1}) = \sup_{\theta \in \Theta_0} p_\theta(y \mid \mathcal{F}_{t-1}) \quad (51)$$

is measurable, positive wherever any p_θ is positive, and finite on the relevant support. This is typically satisfied if Θ_0 is a finite family, or by using an essential supremum over a well-controlled parametric family. Let q_t be a predictable diagnostic subdensity. Define

$$\bar{E}_t = \frac{q_t(Y_t \mid \mathcal{F}_{t-1})}{\bar{p}_0(Y_t \mid \mathcal{F}_{t-1})}. \quad (52)$$

Then for every $\theta \in \Theta_0$,

$$\mathbb{E}_\theta(\bar{E}_t \mid \mathcal{F}_{t-1}) \leq 1, \quad (53)$$

and $\prod_{u \leq t} \bar{E}_u$ is an e-process uniformly valid over the composite null family.

Proof For each fixed $\theta \in \Theta_0$, $p_\theta(y \mid \mathcal{F}_{t-1}) \leq \bar{p}_0(y \mid \mathcal{F}_{t-1})$ pointwise. Hence

$$\mathbb{E}_\theta(\bar{E}_t \mid \mathcal{F}_{t-1}) = \int \frac{q_t(y \mid \mathcal{F}_{t-1})}{\bar{p}_0(y \mid \mathcal{F}_{t-1})} p_\theta(y \mid \mathcal{F}_{t-1}) \mu_t(dy) \quad (54)$$

$$\leq \int q_t(y \mid \mathcal{F}_{t-1}) \mu_t(dy) \quad (55)$$

$$\leq 1. \quad (56)$$

The product result follows from the supermartingale argument. $\square$

Remark 7 (Cost of envelope validity) The envelope denominator is usually not a density and may be much larger than any single fitted predictive density. The resulting e-process can be conservative and computationally difficult for HMMs. This construction is nevertheless useful conceptually because it distinguishes exact fitted-generator validation from uniform validity over a parameter family.

1059 10 Behaviour under misspecification

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1061 The e-process guarantee controls false alarms under the null. Its usefulness under
 1062 alternatives depends on the diagnostic bet. The following results connect e-process
 1063 growth to prequential log-score differences.

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 1068 **Proposition 18** (Conditional log-score decomposition) *Let $p_\star(\cdot \mid \mathcal{F}_{t-1})$ be the true condi-*
 1069 *tional density of Y_t under a data-generating process $P_\star$. Let q_t and p_0 be predictive densities.*
 1070 *If the relevant integrals are finite, then*

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$$\mathbb{E}_\star \left[\log \frac{q_t(Y_t \mid \mathcal{F}_{t-1})}{p_0(Y_t \mid \mathcal{F}_{t-1})} \middle| \mathcal{F}_{t-1} \right] = \text{KL}\{p_\star(\cdot \mid \mathcal{F}_{t-1}), p_0(\cdot \mid \mathcal{F}_{t-1})\} \quad (57)$$

$$- \text{KL}\{p_\star(\cdot \mid \mathcal{F}_{t-1}), q_t(\cdot \mid \mathcal{F}_{t-1})\}. \quad (58)$$

Thus the expected local log-increment is positive exactly when the diagnostic alternative is closer than the null in conditional Kullback-Leibler risk.

Proof By definition,

$$\mathbb{E}_\star \left[\log \frac{q_t(Y_t \mid \mathcal{F}_{t-1})}{p_0(Y_t \mid \mathcal{F}_{t-1})} \middle| \mathcal{F}_{t-1} \right] = \int p_\star(y \mid \mathcal{F}_{t-1}) \log \frac{q_t(y \mid \mathcal{F}_{t-1})}{p_0(y \mid \mathcal{F}_{t-1})} \mu_t(dy) \quad (59)$$

$$= \int p_\star \log \frac{p_\star}{p_0} \mu_t(dy) - \int p_\star \log \frac{p_\star}{q_t} \mu_t(dy), \quad (60)$$

which is the displayed identity. $\square$

Proposition 19 (Asymptotic growth under positive drift) *Let $X_t = \log E_t$ and suppose under $P_\star$ that*

$$\delta_t = \mathbb{E}_\star(X_t \mid \mathcal{F}_{t-1}) \quad (61)$$

exists. Assume the martingale differences $X_t - \delta_t$ satisfy a strong law, for example

$$\frac{1}{T} \sum_{t=1}^T \{X_t - \delta_t\} \rightarrow 0 \quad P_\star\text{-a.s.} \quad (62)$$

If 1105

$$\frac{1}{T} \sum_{t=1}^T \delta_t \rightarrow \delta > 0, \quad (63) \quad \text{1106}$$

then 1108

$$\frac{1}{T} \log E_{1:T} \rightarrow \delta \quad P_{\star}\text{-a.s.} \quad (64) \quad \text{1109}$$

In particular, the e-process grows exponentially at asymptotic rate δ . 1110

Proof Using $\log E_{1:T} = \sum_{t=1}^T X_t$, 1111

$$\frac{1}{T} \log E_{1:T} = \frac{1}{T} \sum_{t=1}^T \delta_t + \frac{1}{T} \sum_{t=1}^T (X_t - \delta_t). \quad (65) \quad \text{1112}$$

The first term converges to δ by assumption and the second to zero by the assumed martingale strong law. 1113

Remark 8 (Power is diagnostic-specific) A large e-process is evidence against the null relative to the alternative used to construct the diagnostic. If the null HMM has an ecologically important defect but the alternative does not predict that defect better, the e-process may not grow. This is a feature rather than a failure: the method localizes model criticism through explicit diagnostic bets. 1114

11 Movement-specific diagnostic catalogue 1115

Table 3 summarizes a practical menu of diagnostics. Each diagnostic can be implemented as a full predictive density ratio, a feature-level e-value, or a blockwise e-value depending on the available null law and the scientific target. 1116

11.1 Diagnostic for the number of states 1117

A natural alternative to a K -state HMM is a $(K + 1)$ -state HMM. Let $p_K(y_t \mid \mathcal{F}_{t-1})$ denote the observable predictive density of the K -state null HMM (ie., p_0) and let 1118

Diagnostic target	Null component	Diagnostic alternative	Ecological interpretation
Number of states	K -state HMM	$K + 1$ state HMM or richer latent structure	Detects behaviour classes merged by the null model
Angular distribution	State-specific von Mises or wrapped distribution	Mixture of von Mises, flexible circular density or circular residual tilt	Detects multimodality, asymmetry or wrong concentration of turning angles
Step-angle dependence	Conditional independence $f_{0,s}^L(l)f_{0,s}^\Theta(\theta)$	Joint circular-linear density or copula on predictive residuals	Detects whether speed and directionality remain dependent within states
State durations	Geometric dwell times implied by HMM	HSMM or duration-aware predictive model	Detects behaviours that persist too long or too briefly for a Markov state process
Temporal nonstationarity	Time-homogeneous parameters	Season-, phase- or covariate-varying diagnostic alternative	Detects periods where the fitted generator stops predicting well
Individual heterogeneity	Shared parameters or random effects in null	Individual-specific or hierarchical diagnostic alternative	Detects held-out animals poorly represented by the fitted population model
Long-horizon spatial behaviour	One-step HMM generator	Blockwise features such as displacement, barrier crossing, residence or space-use summaries	Detects failures visible only through simulated trajectories or aggregated movement behaviour

Table 3 Movement-specific predictive e-diagnostics.

$p_{K+1}(y_t \mid \mathcal{F}_{t-1})$ denote the predictive density of a $(K + 1)$ -state alternative. The ratio

$$E_t^{K+1} = \frac{p_{K+1}(Y_t \mid \mathcal{F}_{t-1})}{p_K(Y_t \mid \mathcal{F}_{t-1})}, \quad (66)$$

where both predictive densities marginalize over their respective latent states. If $\log E_{1:t}^{K+1}$ grows only during certain seasons or individuals, the diagnostic suggests that an additional behavioural mode is needed only locally rather than globally.

11.2 Diagnostic for angular misspecification

Many movement HMMs use a state-specific von Mises distribution for turning angles. A diagnostic alternative can replace each state-specific angular density by a mixture of von Mises distributions, a wrapped distribution with heavier tails, or a feature-level e-value based on circular residuals. The purpose is not necessarily to adopt the

alternative as the final model, but to test whether the null predictive distribution of turning angles can survive a targeted circular bet.

11.3 Diagnostic for residual step-angle dependence

A common HMM assumption is conditional independence of step length and turning angle within state:

$$f_{0,s}(l, \theta) = f_{0,s}^L(l) f_{0,s}^\Theta(\theta). \quad (67)$$

A copula-based alternative uses

$$f_{1,s}(l, \theta) = c_s\{F_{0,s}^L(l), F_{0,s}^\Theta(\theta); \rho_s\} f_{0,s}^L(l) f_{0,s}^\Theta(\theta), \quad (68)$$

with parameters estimated on training data or selected predictably. A feature-level version applies the Rosenblatt residual construction described in Section 8. This diagnostic asks whether, after accounting for latent behaviour, longer steps are associated with different directional persistence or turning behaviour.

11.4 Diagnostic for duration misspecification

A standard HMM implies geometric dwell times in latent states. HSMMs relax this by allowing arbitrary state dwell-time distributions (Langrock and Zucchini, 2011). A duration diagnostic may use an HSMM predictive density as q_t , or a blockwise feature that summarizes likely persistence in behaviours. Because states are latent, duration diagnostics should be based on the predictive density of observations under the duration-aware alternative, not on treating decoded durations as observed without uncertainty.

11.5 Diagnostic for long-term generative behaviour

Some movement failures are not local in $Y_t = (L_t, \Theta_t)$. A model can predict one-step movement well but fail to reproduce utilization distributions, corridors, barriers or returns. The blockwise theory in Theorem 13 supports diagnostics based on multi-step predictive distributions or block features. For example, for a monthly block one may define Z_j as a barrier-crossing count or displacement statistic. A diagnostic alternative $g_{1,j}$ can then be compared with the null block law $g_{0,j}$ through $g_{1,j}(Z_j)/g_{0,j}(Z_j)$.

12 Recommended implementation workflow

A first implementation should prioritize exact split-sample validity and transparent diagnostics.

Algorithm: predictive e-diagnostics for a movement HMM.

1. Split the data into training and validation components. For animal movement, a held-out-individual split is often preferable when several individuals are available.
2. Fit the null HMM M_0 on the training data. Record all preprocessing, covariate choices and decisions regarding the number of states in $\mathcal{H}$.
3. Define a diagnostic menu before validation. Examples include $K + 1$ states, angular flexibility, step-angle dependence, duration structure and blockwise spatial summaries.
4. For each validation time t , compute the null observable predictive density by filtering:

$$p_0(Y_t \mid \mathcal{F}_{t-1}) = \sum_s a_{t|t-1}(s) f_{0,s}(Y_t).$$

5. Compute the diagnostic predictive density or feature density $q_t(Y_t \mid \mathcal{F}_{t-1})$ for each chosen diagnostic.

6. Accumulate	1289
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$\log E_{1:t} = \sum_{u=1}^t \{\log q_u(Y_u \mid \mathcal{F}_{u-1}) - \log p_0(Y_u \mid \mathcal{F}_{u-1})\}.$	1292
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	1295
7. Monitor the anytime-valid threshold $\log(1/\alpha)$. For $\alpha = 0.05$, this threshold is	1296
$\log(20) \approx 2.996$.	1297
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8. Report both global and localized evidence: individual e-processes, soft state-	1299
localized processes, segment-specific processes and local increments.	1300
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9. Interpret the result comparatively. A large e-process indicates that the diagnostic	1302
alternative predicted the validation sequence better than the fitted HMM in a	1303
sequentially valid sense; it does not prove that the diagnostic alternative is the	1304
true ecological mechanism.	1305
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13 Simulation study	1311
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The simulations were designed to check whether the theoretical guarantees translate	1314
into a reproducible R workflow and whether targeted diagnostic alternatives accumu-	1315
late evidence under controlled movement-model failures. All scenarios use simulated	1316
movement trajectories with step lengths and turning angles, forward filtering for the	1317
observable HMM predictive density and cumulative log e-processes monitored against	1318
thresholds $\log(1/\alpha)$. The scenarios are not meant to form a benchmark of HMM-fitting	1319
algorithms. They are controlled checks of the diagnostic objects introduced above.	1320
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The main article reports seven scenarios. Scenario S1 checks calibration under	1326
a correctly specified fixed HMM. Scenarios S3 to S6 evaluate targeted diagnos-	1327
tic power for an insufficient number of states, angular misspecification, residual	1328
step-angle dependence and non-geometric duration structure. Scenario S7 studies pre-	1329
dictable localization of a temporally and state-localized failure. Scenario S8 evaluates	1330
predictable mixtures and switching among diagnostics.	1331
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Sc.	Null gen. p_0	Data generator	Diagnostic q_t	K_0	K_{true}	T	n_{rep}	Threshold	Signal rate (SE)
S1	2-state HMM	2-state HMM	Exact 2-state	2	2	300	300	$\log(20)$	0.030 (0.010)
S3	2-state HMM	3-state HMM	$K + 1$ states	2	3	300	300	$\log(20)$	0.997 (0.003)
S4	2-state HMM	2-state (misspec. angle)	Diffuse angle	2	2	300	300	$\log(20)$	1.000 (0.000)
S5	2-state HMM	2-state (copula dep.)	Joint copula	2	2	300	300	$\log(20)$	1.000 (0.000)
S6	2-state HMM	2-state HSMM	Duration-aware	2	2	500	300	Blockwise	0.000 (0.000); 1.000 (0.000)
S7	2-state HMM	2-state (localized fail.)	State-localized	2	2	300	300	$\log(20)$	0.063 (0.014); 0.900 (0.017)
S8	2-state HMM	2-state (change point)	Switching mix	2	2	300	300	$\log(20)$	0.036 (0.011); 1.000 (0.000)

Table 4 Simulation scenarios retained in the main article. For scenarios with two entries in the signal-rate column, the first value refers to the null calibration run and the second to the targeted alternative. For S6 specifically, the first value is the crossing rate under an HMM null calibration run, and the second is the crossing rate under the HSMM alternative. With 300 replicates, a signal rate of 0.000 implies an approximate upper 95% confidence bound of about 0.01.

The supplementary material contains five scenarios that are useful but less central to the main narrative. Scenario S2 studies a train/validation workflow with oracle-state parameter estimation and is therefore treated as a sensitivity analysis rather than a main HMM-fitting result. Scenario S9 demonstrates why parallel products of diagnostics are not valid by default. Scenario S10 studies a longer-horizon blockwise straightness diagnostic. Scenario S11 studies validation by individuals and cross-fitted averages. Scenario S12 studies a conservative finite-family composite-null envelope.

Table 4 gives the main numerical summary. Under the correctly specified null in S1, the crossing rate at $\alpha = 0.05$ is 0.030, consistent with conservative finite-sample behaviour. The targeted alternatives in S3 to S5 have high power in the controlled alternatives, with crossing rates of 0.997, 1.000 and 1.000, respectively. The duration scenario S6 gives no false signals under the HMM null and detects the HSMM-like alternative in every replicate. The localization scenario S7 shows why local increments should be reported: the global diagnostic crosses in 0.063 of replicates at $\alpha = 0.05$, whereas the diagnostic localized to the failure window and predicted state crosses in 0.900 of replicates. Finally, S8 confirms that predictable mixtures and switching can be used without the severe false-signal inflation associated with non-valid parallel

products. Together, these scenarios check calibration, power, specificity, localization
and diagnostic combination in separate controlled settings.

These simulations are not intended to exhaustively compare HMM estimators or
movement models. Their purpose is narrower: to verify that the proposed e-diagnostic
objects can be computed reproducibly, that the null behaviour is compatible with
the theory in controlled settings and that each diagnostic responds to the failure it
is designed to target. The supplementary scenarios document additional safeguards
and extensions that are useful for implementation but would broaden the main article
beyond its central message.

13.1 Comparison with Classical Pseudo-Residual Diagnostics

To evaluate the practical utility and power of the proposed predictive e-diagnostics,
we compare them against classical diagnostic tests based on pseudo-residuals. It is
important to distinguish marginal probability integral transform (PIT) residuals from
joint sequential Rosenblatt residuals. Under a correctly specified movement HMM,
marginal PIT residuals are marginally Uniform(0,1) variables, but they are gener-
ally not independent because the observable predictive mixture induces dependence
through state uncertainty. True joint sequential Rosenblatt residuals, constructed con-
ditionally as described in Section 8.1, are independent Uniform(0,1) variables, which
can be mapped to standard normal variables (Zucchini et al., 2016).

- **Kolmogorov-Smirnov (KS) Test:** Applied to the circular PIT turning angle
residuals ($u_{t,\text{angle}}$) and marginal step length PIT residuals ($u_{t,\text{step}}$) to test for
deviations from uniformity.
- **Ljung-Box Test:** Applied to the normal step pseudo-residuals ($z_{t,\text{step}}$) at lag 5
to check for residual autocorrelation.

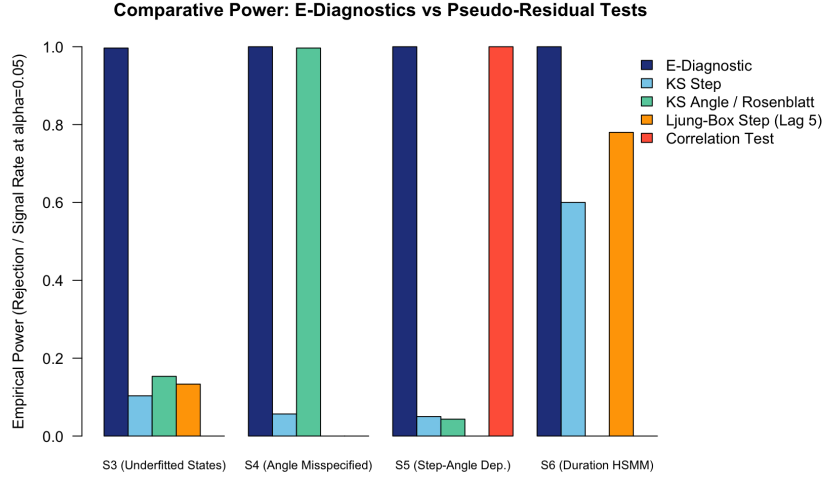


Fig. 1 Signal/rejection rates in scenarios S3–S6. All scenarios use 300 replicates; $T = 300$ for S3–S5 and $T = 500$ for S6.

- **Correlation Test:** Applied to the normal step and angle pseudo-residuals to test for residual joint dependence (specifically in S5, using the Rosenblatt transform of angles conditional on steps).

Figure 1 summarizes the empirical signal and rejection rates across 300 simulated trajectories, using the scenario-specific validation lengths reported in Table 4.

The comparison highlights several key advantages of the e-diagnostic framework:

1. **Underfitted States (S3):** The e-diagnostic detects the missing third state in 99.7% of the replicates. In contrast, the classical marginal KS tests on steps and angles fail to reject the null in most replicates (10.3% and 15.3% power, respectively), and the Ljung-Box autocorrelation test achieves only 13.3% power.
2. **Angular Misspecification (S4):** Both the angular e-diagnostic and the circular-linear KS test on turning angles exhibit near-perfect power (100% vs. 99.7%).

3. **Step-Angle Joint Dependence (S5):** The copula e-diagnostic detects the residual dependence in 100% of the replicates. The marginal KS tests fail completely (5.0% and 4.3% power) because they are designed to detect univariate deviations only. While the targeted correlation test on the joint normal residuals also achieves 100% power, the e-diagnostic achieves this without sacrificing sequential validity and allows for online monitoring.
4. **Non-Geometric State Durations (S6):** The blockwise duration e-diagnostic achieves 100% power in detecting the HSMM-like state transitions, whereas the step-length KS test and Ljung-Box test reach only 60.0% and 78.0% power, respectively.

These results demonstrate that e-diagnostics are highly sensitive to structured model misspecifications while preserving anytime-validity. However, this comparison primarily illustrates targeted power under specific, controlled scenarios. It does not imply a universal dominance of e-diagnostics over classical tests, especially since the classical tests reported here are evaluated retrospectively without adjustments for sequential monitoring or multiple testing.

14 Elk movement application

As a real-data application, we used the elk movement data distributed as `elk_data` in the R package `moveHMM` (Michelot et al., 2016). The data are associated with the elk movement study of Morales et al. (2004). The dataset contains four individuals and 735 locations with projected coordinates and distance to water. We used `moveHMM::prepData` to compute step lengths and turning angles. We implemented a Leave-One-Animal-Out (LOAO) cross-validation protocol: for each of the four individuals in turn, we held it out for validation and trained the candidate models on the remaining three individuals. Figure 2 displays the trajectories of the four individuals.

Trajectories of the four elk individuals

Under Leave-One-Animal-Out cross-validation, each individual is used for validation in turn.



Fig. 2 Trajectories of the four elk individuals. Under the Leave-One-Animal-Out (LOAO) protocol, each trajectory is used for validation in turn.

For each of the four folds, we fitted HMMs with $K = 2, 3, 4$ states on the training individuals only, using Gamma step lengths and von Mises turning angles with no transition covariates. To handle local optima, models were fitted using `moveHMM::fitHMM` with 10 random perturbations of the initial parameters. The state distribution at the first observation was fixed to the stationary distribution. Label switching between cross-validation folds was resolved by ordering the states by their mean step lengths before prediction. The fixed mixture diagnostic allocated equal weight ($1/2$) to the $K = 4$ density and the diffuse-angle density. Missing first steps or turning angles due to missing coordinates were dropped by `prepData`, and the filter was explicitly restarted for each individual to ensure valid prediction. Initial parameters for the 10 random perturbations were drawn uniformly between realistic bounds based on the sample quantiles of the data. The diffuse-angle diagnostic retains the fitted state weights and

Model selection across Leave-One-Animal-Out folds

BIC consistently selects $K = 3$ as the null model for all training sets.

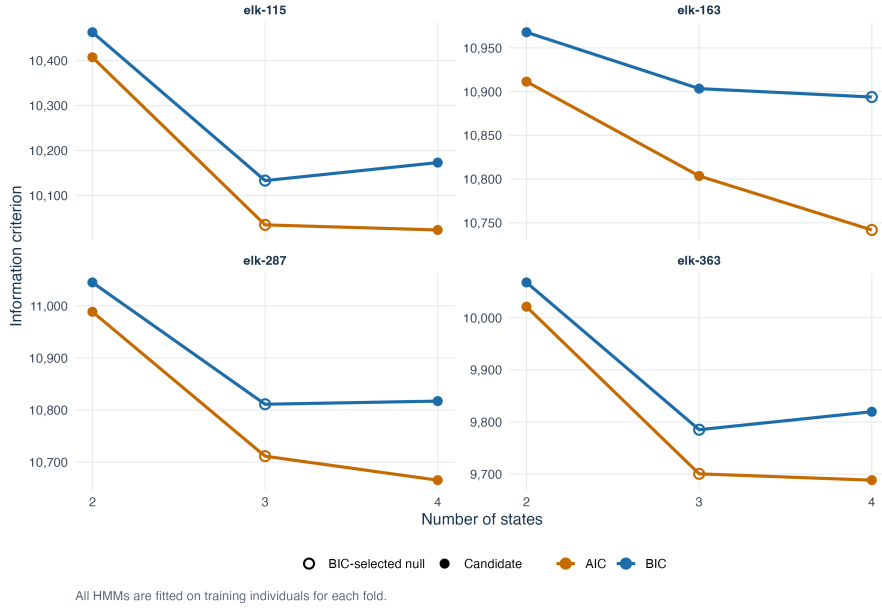


Fig. 3 AIC and BIC for the fitted HMMs across the four training folds. BIC selects $K = 3$ for three folds and $K = 4$ for one fold.

step length distributions, modifying only the turning angle density to a more diffuse von Mises distribution. All diagnostic choices were fixed before observing the validation trajectories. Figure 3 and Table 5 report the model-selection metrics. BIC selects $K = 3$ as the null model for three folds (elk-115, elk-287, and elk-363) and $K = 4$ for fold elk-163. To maintain consistency and allow population-level cross-fitted pooling, we fix the null generator to $K = 3$ states and the diagnostic alternative to $K = 4$ states across all validation individuals.

We then evaluated four diagnostics on each validation individual: a full-density $K + 1$ diagnostic comparing the $K = 3$ null HMM with the $K = 4$ alternative HMM, a diffuse-angle diagnostic obtained by halving the fitted angular concentrations under the null, a fixed mixture of these two diagnostic densities (allocating equal weight to each), and a state-localized version of the $K + 1$ diagnostic using the filtered predictive probability of the long-step state. Figure 4 shows the resulting cumulative paths for

	Fold (Validation Individ.)	States (K)	Negative Log-Likelihood	Parameters	AIC	BIC
	elk-115	2	5190.649	13	10407.298	10462.919
	(Training $n = 533$)	3	4994.368	23	10034.736	10133.142*
		4	4976.707	35	10023.414	10173.162
	elk-163	2	5442.706	13	10911.412	10967.860
	(Training $n = 568$)	3	5378.781	23	10803.562	10903.430
		4	5335.920	35	10741.841	10893.815*
	elk-287	2	5481.246	13	10988.493	11044.826
	(Training $n = 563$)	3	5332.742	23	10711.484	10811.149*
		4	5297.755	35	10665.510	10817.175
	elk-363	2	4999.619	11	10021.237	10067.837
	(Training $n = 511$)	3	4830.232	20	9700.463	9785.190*
		4	4813.181	31	9688.363	9819.690

Table 5 Model selection metrics for fitted HMMs across Leave-One-Animal-Out training sets. The * symbol denotes the model selected by BIC for each fold.

Leave-One-Animal-Out cumulative log e-processes

The dashed line is the $\alpha = 0.05$ threshold, $\log(20)$. Tracks vary across individual animals.

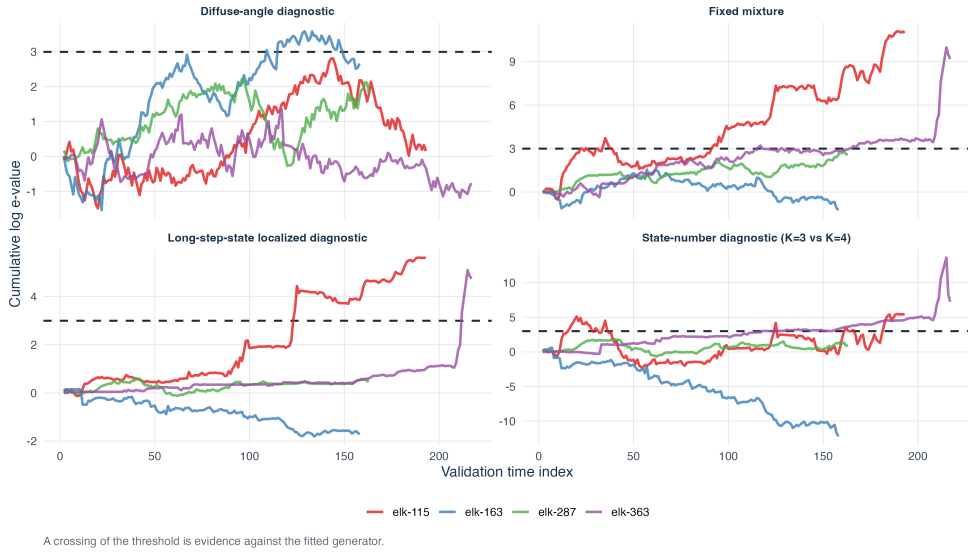


Fig. 4 Cumulative log e-processes for the four validation elk individuals across the four diagnostics. The dashed horizontal line is the $\alpha = 0.05$ threshold, $\log(20) = 2.996$.

the four individuals, and Table 6 summarizes their crossings, maxima, and final values, alongside the population-average cross-fitted e-values (Proposition 16).

The diagnostics show substantial variation across individuals. For elk-115 and elk-363, the $K + 1$ state diagnostic and the fixed mixture cross the $\alpha = 0.05$ threshold ($\log(20) = 2.996$), suggesting that the three-state null HMM is inadequate for

Validation Individual / Diagnostic	Signal	Crossing Time	Max log e	Final log e
elk-115 ($n = 192$)				
$K = 3$ vs $K = 4$	Yes	16	5.459	5.413
Diffuse angle	No	NA	2.811	0.154
Fixed mixture	Yes	22	11.089	10.981
Localized $K + 1$	Yes	123	5.618	5.605
elk-163 ($n = 157$)				
$K = 3$ vs $K = 4$	No	NA	0.610	-12.216
Diffuse angle	Yes	109	3.585	2.663
Fixed mixture	No	NA	1.556	-1.244
Localized $K + 1$	No	NA	0.165	-1.725
elk-287 ($n = 162$)				
$K = 3$ vs $K = 4$	No	NA	1.854	1.056
Diffuse angle	No	NA	2.220	1.801
Fixed mixture	No	NA	2.836	2.592
Localized $K + 1$	No	NA	0.619	0.513
elk-363 ($n = 214$)				
$K = 3$ vs $K = 4$	Yes	118	13.627	7.246
Diffuse angle	No	NA	1.337	-0.761
Fixed mixture	Yes	116	9.973	9.174
Localized $K + 1$	Yes	212	5.095	4.734
Population Average ($N = 4, n_{tot} = 725$)				
$K = 3$ vs $K = 4$	Yes	NA	6.009	6.009
Diffuse angle	No	NA	1.706	1.706
Fixed mixture	Yes	NA	9.747	9.747
Localized $K + 1$	Yes	NA	4.573	4.573

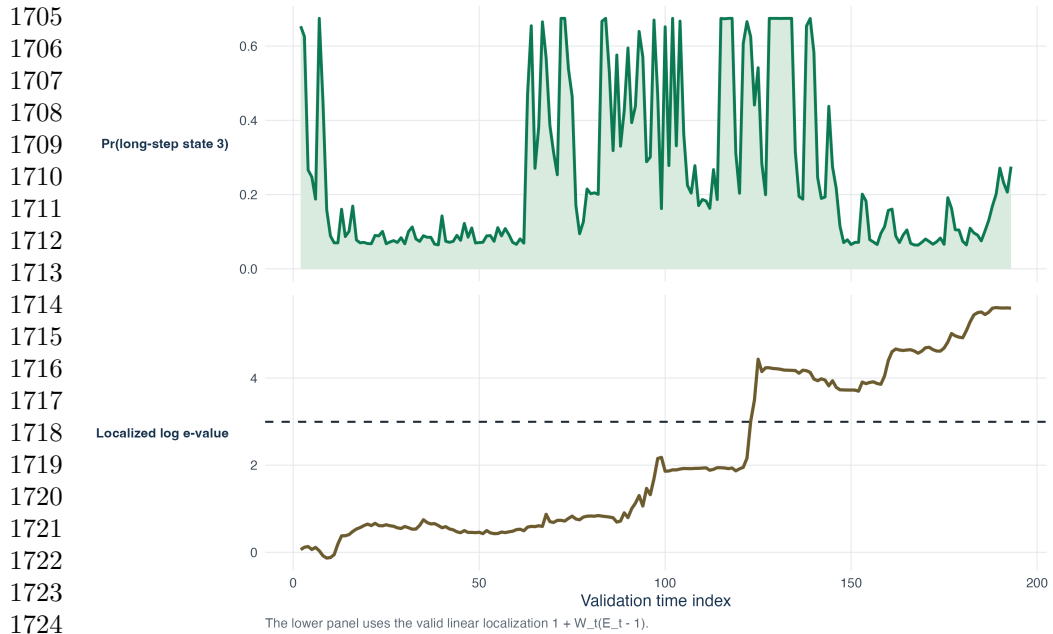
Table 6 Leave-One-Animal-Out predictive e-diagnostics for the four elk individuals and the population-average cross-fitted e-values. The threshold for $\alpha = 0.05$ is $\log(20) = 2.996$. For the population average, the final log e-value represents $\log E^{\text{cf}}$ and the max log e-value is identical to the final value by definition.

these trajectories. The state-localized diagnostic also crosses for these two individuals, supporting the interpretation that the predictive discrepancy is concentrated during periods assigned high probability of the long-step state. For **elk-163**, the diffuse-angle diagnostic crosses the threshold, suggesting that the diffuse-angle diagnostic predicts this trajectory better than the fitted angular component for this animal, while the diagnostic for the number of states does not cross. For **elk-287**, none of the individual diagnostics cross the threshold.

Importantly, Proposition 16 allows us to pool evidence across individuals using the population-average cross-fitted e-value, $E^{\text{cf}} = \frac{1}{N} \sum_{i=1}^N E^{(i)}$. Under the LOAO composite null hypothesis H_0^{LOAO} , which posits that each validation trajectory $Y^{(i)}$ is generated by the HMM fitted on the remaining $N - 1$ individuals, E^{cf} remains a valid terminal e-value. As shown in Table 6 and Figure 6, the population-average log e-value exceeds the terminal e-value threshold for the diagnostic for the number of states

1703 The localized signal waits for long-step conditions (elk-115)

1704 Filtering supplies predictable state weights before each validation observation.



1725 **Fig. 5** Predictable state-localized diagnostic for validation individual **elk-115**. The upper panel
1726 shows the filtered predictive probability assigned to the long-step state before each validation obser-
1727 vation. The lower panel shows the corresponding localized cumulative log e-process.

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1733 ($\log E^{cf} = 6.009$), the fixed mixture ($\log E^{cf} = 9.747$), and the localized diagnostic

1734 ($\log E^{cf} = 4.573$). Only the diffuse-angle diagnostic does not exceed the terminal

1735 threshold at the population level ($\log E^{cf} = 1.706$). In this illustrative dataset, the

1736 LOAO diagnostics suggest that the three-state fitted generator is inadequate for some

1737 held-out individuals, with population-average evidence concentrated in the number-

1738 of-states and long-step-localized diagnostics. Figure 5 shows the predictable state

1739 weights and localized cumulative e-process for **elk-115**.

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1744 This application illustrates the Leave-One-Animal-Out cross-fitted workflow: train

1745 the HMM on a subset of individuals, specify diagnostic alternatives, evaluate the e-

1746 processes on held-out animals, and compute population-level cross-fitted e-values to

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Summary of final log e-values across individuals

Individual results (circles) and population cross-fitted average (green diamond).

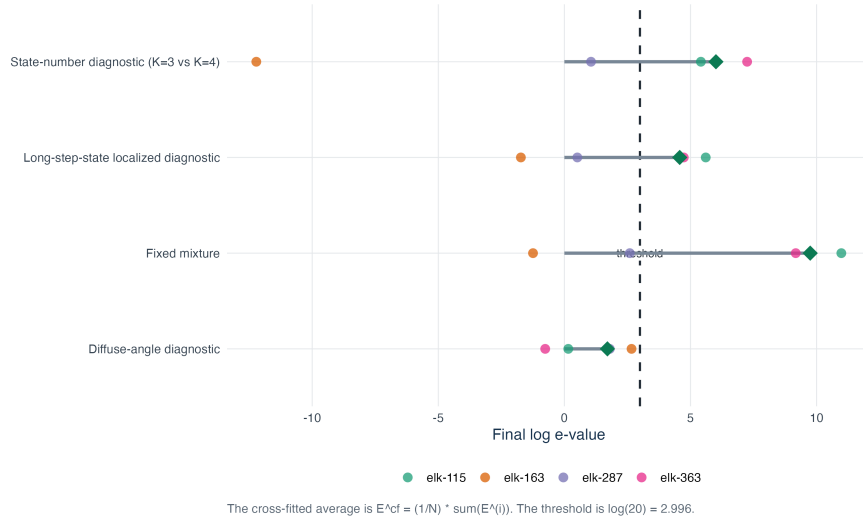


Fig. 6 Summary of final log e-values across validation individuals (circles) and the population-average cross-fitted e-value (green diamond). The dashed line represents the $\alpha = 0.05$ threshold, $\log(20) = 2.996$.

pool evidence. The results illustrate how predictive e-diagnostics can reveal individual-specific lack of predictive fit and can be combined into a population-level terminal e-value.

15 Discussion

This theoretical framework turns model checking for multi-state movement models into a sequence of explicit predictive bets. The null HMM is assessed as a sequential generator of held-out movement data. The latent states are integrated out through filtering. Diagnostic alternatives define targeted predictive competitors. The resulting cumulative ratios are e-processes under the fixed train/validation null, giving anytime-valid thresholds for sequential model criticism.

The framework adds several elements that are specific to movement HMMs. First, it prevents a common conceptual error: e-validity is based on the observable predictive density, not on a density conditional on decoded states. Second, it provides a

way to localize model failure using predictable weights, including habitat indicators and filtered state probabilities. Third, it supports diagnostics tailored to movement structure: turning-angle misspecification, residual circular-linear dependence, non-geometric dwell times and long-horizon spatial behaviour. Fourth, it clarifies how several diagnostics can be combined. Predictable switching and mixtures are valid; parallel products on the same validation observations are not valid by default.

The main limitation is the treatment of fitted parameters. The exact guarantee is cleanest when the fitted null and diagnostic alternatives are fixed before validation. This is a strength for a first methodological article because the null statement is transparent: conditional on training, does the fitted HMM generate the validation trajectory credibly? Same-sample exploratory workflows are still useful but require caution. The supplementary simulations therefore treat oracle-state parameter estimation, validation by individuals and composite-null envelopes as sensitivity analyses or extensions rather than as the main empirical evidence.

The real-data example illustrates both the promise and the caution required in practice. Predictive e-diagnostics can reveal that a fitted generator performs poorly for a held-out animal, and localized versions help identify the part of the fitted behavioural structure associated with the signal. At the same time, ecological interpretation requires additional work: the diagnostic menu, train/validation split and model-selection rule should be pre-specified for confirmatory use, and the results should be replicated across datasets or validation splits before drawing definitive biological conclusions.

Code and Data Availability

The elk data used in the application are publicly available in the supplementary materials of [Michelot et al. \(2016\)](#) and via the `moveHMM` package. The repository contains R scripts, simulation parameters, random seeds, and a README to fully

reproduce the tables and figures. While generated outputs are ignored by Git as noted in the supplement, the verifiable results and full scripts can be accessed at https://github.com/AurelienNicosiaULaval/predictive_e_diagnostics_hmm.

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